

MANUFACTURING AUDIT PROGRAM #54-00-00

The release of this audit program to parties external to IBM requires the written approval of the Chief Auditor.

This program is intended as a guide for assessing risk management in this process through the performance of an audit or a similar engagement. Risk management is the process to identify, manage, and control potential events in order to provide reasonable assurance related to the achievement of IBM's business and control objectives. IBM's control objectives fall into five broad categories:

1. compliance with external laws and regulations, as well as IBM policies and procedures
2. accomplishment of business objectives
3. reliable information for financial reporting and decision making
4. efficient operations, and
5. safeguarding both intellectual and physical assets.

The control assessment questions are intended to address process objectives from the categories above and the risks to achieving those objectives. Objectives and risks in the actual process under review may differ based on organizational, operational, regulatory, and economic factors. As such, individuals performing the engagement must formulate a test strategy based on risks relative to the actual process under review, regardless of whether the risks are identified within this program.

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DATE CHANGED	SECTION #	SUBTITLE IN SECTION	CHANGED BY (NAME)	COMMENTS
Dec 2025	Self-Service Reports		Misook Cha	Delete this temporarily until audit can get complete access to extract the secondary control in Open Page
Dec 2025	Process and Objective, Planning: Reference Material, Test Section	SE004.P, AC001-003.P, AC005.P, EC001.P	Misook Cha	Minor word update, Add full name of acronym Delete one testing attribute related with F&O163 as it was agreed with GARS audit.
Dec 2025	Test Section	AC005.P	Misook Cha	Add detail / additional testing strategy
June 2025	TEST Sections	SE004.P, EC001.P, IC001.P	Misook Cha	Update minor wordings per the latest process in the Test Approach
June 2025	PLANNING: Audit Program Risk Alignment		Misook Cha	Specify individual tests in RAE table accordingly.
June 2025	PLANNING: Audit Program Risk Alignment TEST Sections		Misook Cha	Change the Primary Test to Secondary Test on SC001, SC002, SA001, SA002, HS001, HS002, HS003
June 2025	PLANNING: Data Analysis TEST Sections		Misook Cha	Update the Data Analysis from "Data mining" in all test sections Removed IDR, Data mining, Reference Material remark in each test for simplification & avoiding duplicated instructions.
March 2025	Planning: Reference Material		Misook Cha	Updated the reference per the latest address
March 2025	4. Inventory and Asset Controls		Misook Cha	Independent Inventory counting testing approach was updated
March 2025	11. Environmental Management	Professional Self-Assessment (EM001.P)	Misook Cha	Remove the control point as 1) the testing is duplicated with EM023.P in the audit program 47-10-00 and 2) Hungary VAC location which was specifically designed for this control point was closed.
November 2023	Planning: Data Analysis		Jan Aldrin Pe	Added notes for Self-Service Reports

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August 2023	Supply Chain Execution	Material Requirements Planning (SE001.P) Supply Order Management (SE002.P) Manual Pulls (SE003.P) Excess, Surplus and Scrap (SE004.P)	Jan Aldrin Pe	Updated tests
August 2023	Manufacturing Operations	Customer Order Management (MO001.P)	Jan Aldrin Pe	Added test from Customer Availability
August 2023	Inventory and Asset Controls	Physical Security (AC001.P) Sensitive Parts Management (AC002.P) Inventory Scrap (AC004.P) Product and Inventory Returns (AC005.P)	Jan Aldrin Pe	Updated tests
August 2023	Customer Satisfaction	Inventory Controls and Verification (AC003.P)	Jan Aldrin Pe	Deleted entire section
August 2023	System Access and Separation of Duties	Separation of Duties (SA002.P)	Jan Aldrin Pe	Updated tests
August 2023	Supplier Payment Controls	PO Processing (SP001.P) Supplier Payment Controls (SP002.P)	Jan Aldrin Pe	Updated tests Deleted entire sub section

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August 2023	Healthy and Safety Programs	Equipment / Process Installation (HS001.P) Hazardous Work Activities (HS002.P)	Jan Aldrin Pe	Updated tests
August 2023	Export Compliance Audit Program Risk Alignment	Regulatory Licenses Requirements (EC001.P)	Jan Aldrin Pe	Updated based on changes and included Export and Import Compliance to Supply Chain Execution
February 2023	All	All	CA&AS	Converted from Notes into a Word document and into a Box folder

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PROCESS AND OBJECTIVE

For purposes of this audit program, hardware manufacturing is defined as the entire process for ensuring customer requirements are met while optimizing costs, accurately accounting for inventory, maximizing revenue, and protecting IBM assets.

The process begins by developing an annual inventory plan based upon expected customer demands. This plan should utilize product profiles that establish ordering parameters for all major products. These parameters would include items such as frozen zones, minimum order quantities, cycle time, protective stock, supplier lead times, etc.

Comprehensive measurements provide management visibility to inventory performance against committed inventory targets. A management system should be in place to accommodate customer demand fluctuations and ensure that inventory levels remain appropriate, and costs are minimized.

A process for ensuring inventory financial records is accurate should exist. Inventory verification should be performed for all portions of inventory, i.e., physical inventory, work-in-process (WIP), finished goods, returned inventory, defective inventory, and consignment inventory.

Security measures should be adequate to protect all inventory assets from loss or theft.

Note: For operations that include a Customer Solution Center (CSC) apply the Control Assessment Questions located in section 9. Customer Solution Center.

Overall, the purpose of hardware manufacturing is to maximize revenue and profit and improve the ease of doing business with IBM. This is accomplished by meeting customer quantity, quality and timeliness demands for all inventories. While, at the same time, maintaining optimal inventory levels, minimizing disruptions to suppliers, minimizing inventory write-offs, maintaining financial record integrity and effectively safeguarding IBM's assets.

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PLANNING: Audit Program Risk Alignment

Manufacturing

Defined Risk Area	Audit Risk Level	Key Control Question	Test Number
Supply Chain Execution	High	Supply Chain Execution, Manufacturing Operations, Health and Safety Programs, Export Compliance, Import Compliance	SE001.P, SE002.P, SE003.P, SE004.P, SE005.P MO001.P, HS001.S, HS002.S, HS003.S EC001.P, IC001.P,
Inventory and Asset Controls	Medium	Inventory and Asset Controls Property Control	AC001.P, AC002.P, AC003.P, AC004.P, AC005.P PC001.P
Supplier Controls	Medium	Measurements, Supplier Payment Controls	MM001.P, SP001.P, SP002.P
Supply Chain Planning	Medium	Supply Chain Planning	SC001.S, SC002.S
System Access and Separation of Duties	Low	System Access and Separation of Duties	SA001.S, SA002.S

*Create an additional risk area if issues are observed in relation to customer solution centers.

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PLANNING: Data Analysis

Data analysis should be a part of every engagement where data is available in databases or data warehouses that can be accessed through query programs. This concept enhances the risk-based auditing approach, makes the audit process more efficient, and increases the productivity of the audit team. When the audit team does not have the skills required to perform these tasks, assistance should be sought from the Technology team.

All data used for analysis must be reconciled to an independent source so that the auditor has confidence that the universe is complete.

The [Data Analysis engagement procedures in the Audit Manual](#) (link internal to CA&AS members only) provide further suggestions.

This table shows what data can be reviewed and analysis techniques.

Data to be Analyzed	Data Location	Analysis to be Performed	Possible Anomalies	Concern
Customer shipments	Shipping and logistics application: Most common applications: SAP, MAPICS, RPDS	Compare ship date to PSSD date.	1. Late shipment. 2. Early shipment. 3. Missed shipment	1a. Impact to customer satisfaction. 1b. Missed revenue. 1c. Premium freight charges. 2a. Revenue claimed inappropriately.
Inventory adjustments	Logistic application. Most common applications: SAP, MAPICS	Identify re-occurring adjustments for the same part number.	1. Lack of control of assets.	1a. Possible theft.
Inventory adjustments	Logistic system: SAP, MAPICS	Timing of adjustments prior to API / RIA.	1. Manipulation of metric results.	1a. Metric results would not be a true reflection of control environment.
Receipts	Compare the amount of time it takes from receipt of shipment from carrier until the product is available for manufacturing use.	Correct receiving scan	1. Impact to OTS 2. Expedited orders and premium freight charges due to slow receipt.	
List of sensitive parts variance		Review if unexplained negative variances exist to determine a pattern.		Possible theft.
List of Manual Pull & Replenishment Pull		Review the list for reoccurring supplier, part number, date, or parts analyzer		

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PLANNING: Data Analysis (Continued)

Data to be Analyzed	Data Location	Analysis to be Performed	Possible Anomalies	Concern
List of unplanned / manual requisitions and emergency orders		Validate that: - Manual requisitions were approved - The orders were justified check if parts numbers were in the list of reserved parts There is a trend per parts numbers, PO, parts analyzer		Ineffective customer order management could result in missed customer shipments, customer dissatisfaction, and additional costs.

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PLANNING: Reference Material

Planning Matrix/Scope Guide

- The IBM Manufacturing mission is evolving. Distribution operations, receiving, and shipping are now Global Logistics responsibilities and have been outsourced to LLP suppliers. Cost Accounting has been centralized and is no longer a site responsibility. Segments of manufacturing within a site have been outsourced. The rate of change varies by site. In addition, not all processes are executed at every site. Some processes are only executed at control tower sites.
- The following matrix should be used as a guide when determining the scope areas to be reviewed when planning for an audit engagement for in-house manufacturing including: inventory management/distribution operations/cost accounting/procurement (if planned to be concurrent). During the preplanning review with the organizations management team, complete the following preplanning matrix guide.

Test Description	Global or Local	Site Level?	Outsourced or Other Org?
Inventory Planning	G + L		
Excess, Surplus & Scrap (ESZ) Management	Automated	Hubs	
Engineering Change (EC) Management			
Supply / Demand Management System	Global Systems		
Material Requirements Planning (MRP) and Supply Order Management			
Supply Chain Specialist			
Replenishment			
Receiving			
Shipping			
Customer Order Management			
Rapid Response			
Customer Supply Availability			
Customer Satisfaction and Surveys			
Asset Protection			
Sensitive Parts Management			
Physical Security			
Inventory Controls			
Parts Movement			
Rotating Inventory Counts (RIC)			
Rotating Inventory Audits			
Annual Physical Inventory			
Returned Parts Control			
Return Parts Miscellaneous Equipment Specifications (RPMES) Management			
Product Return i.e. shipped and uninstalled			
Consigned good return including vendor re-work			
Scrap			
Separation of Duties			

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PLANNING: Reference Material (Continued)

Asset Protection

- All below guidelines can be downloaded on the following site:
<https://w3.ibm.com/w3publisher/asset-protection>
 - CP 10.12 Asset Protection – Excess & Surplus Parts Inventories
 - CP 10.13 Asset Protection – Normal and Sensitive Parts Security
- CP 10.15 Asset Protection – Returned Parts MES and EC Process Requirements

Inventory Cost and Verification

- AI-INVCA00-005 Rotating Inventory Audits of Inventories Conducted by Accounting:
<https://ibm.ent.box.com/s/lixo7vvsi3u2b319rzg33985zli3eu79>
- AP INV-VER- Inventory Verification: <https://ibm.ent.box.com/s/zzvtthvvv8h3vamukwccslndbg7q91r>
- AP INV-CON- Consignment Inventory Accounting and Controls:
<https://ibm.ent.box.com/s/q566sy0guxwfnr94sk8bc0otpj2malre>
- AP INV-2-COST - Inventorying Cost for New Technology Product and System Brands:
<https://ibm.ent.box.com/s/ux3t2yxljoyjcnqwn5a741avi6d4shs>

Labelling/ Receiving/Shipping

- CB03700000 Tamper Evident Packaging:
<https://ibm.ent.box.com/s/s7bzbmjsqfkmxcousnm9ijuds3rvkmu9>

For below two standards, please visit <https://w3.ibm.com/w3publisher/ibm-internal-standards> and select the file per the directory described as below.

- Internal Standard C-S 1-1121-015: Automatic Identification (AI) for Packaging, Distribution and Manufacturing; Bar Coded Labels:
 - Open Drive Link to IBM current Standards by DCS number (All IBMers can view and download -> W3-Box standard's Procedures review -> Download "e-SiS w3-Box Tip – IBM stds C-S 1-121-015.pptx")
- Corporate Bulletin C-B 0-3700-000: Packaging and Sealing of IBM or Customer Assets:
 - Open Drive Link to IBM current Standards by DCS number (All IBMers can view and download -> 0-Documents and Procedures -> 0-3000 -> C-B 0-3700-000 – 12 – 1997)

Scrap

- IBM99-001 Accounting for Plant Inventory Reserves:
<https://ibm.ent.box.com/s/9lchfr1ci9l9ltpyvt1trInt1rbyw2gp>
- Scrap Process: <https://w3.ibm.com/w3publisher/global-fixed-asset-management-wiki/my-asset/asset-disposal/scrap>

System Access and SOD

- IT Security Standard (ITSS): Please refer to the section 4.0 Access Controls and 6.0 Operations Security in IT Security Standard (ITSS) managed by IBM Cybersecurity – PSG.
<https://pages.github.ibm.com/ciso-psg/main/standards/itss.html>
- Fin 182 Management of Self-Assessment/Appendix C SOD:
<https://w3.ibm.com/w3publisher/financebc/ibm-framework/ci-fin-182-appendix-c>

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PLANNING: Protocol for Auditing

Protocol for Auditing in an Outsourced Environment

In support of the outsourced contractual relationship post-divestiture; the following audit protocol is to be implemented within the Audit Programs listed below:

- GL - Physical Logistics, 32-10-00
- Parts Logistics, 32-20-00
- GL - Trade Compliance, 32-30-00
- Manufacturing, 54-00-00

IBM/Logistics Supplier Audit Protocol

This protocol is to cover formal reviews, inspections, and or audits with the intent to provide both companies with a framework to conduct these engagements in an efficient and productive manner.

These engagements are conducted with the intent to determine the controls health of key processes and to implement process improvements to assure and sustain a strong controls posture. This is beneficial to both companies to meet an environment of increasing regulatory and legal requirements.

In some cases, Lead Logistics Providers (LLPs) use and manage a network of suppliers that are contracted to IBM. In other cases, described as LLP below, the LLP uses and manages their own network of suppliers. LLP suppliers have assumed direct responsibility for IBM's physical logistics needs, becoming IBM's Fourth-Party Logistics (4PL) Suppliers. LLP suppliers have also assumed the management of IBM's existing global logistics contracts which are now Third-Party Logistics (3PL) suppliers of logistics activities to IBM. While LLP suppliers are IBM's partners, they are separate legal entities, not owned by IBM.

All contracts with service providers where IBM Global Logistics Procurement was previously the buyer have been transferred to LLP suppliers (few exceptions only). IBM Global Logistics no longer has a direct contract responsibility with LLP sub-tier suppliers. The contract responsibility with the 3PL service providers is now owned and managed by LLP suppliers.

Primary point of contact for reviews, inspections, and/or audits is with the IBM Control Tower representative.

The standard notification from IBM to LLP suppliers will be 30 days in advance of review. There will be circumstances; however, where the schedule will be dictated by other factors, such as preparation for a short notice customs audit, resulting in less than 30-day notification.

Establish a LLP supplier point of contact for specific engagement. LLP supplier will assign a company representative to speak on behalf of the company in discussions. When requests fall into LLP area of responsibility the IBM Control Tower representative will involve the designated GSCO interface.

As a practice, requests of LLP suppliers or sub-tier suppliers should be focused on results and not with the supplier owned processes, procedures or work instructions. Because of IBM's regulatory and legal risks/liability, requests for data and observation of LLP supplier's or sub-tier supplier's security measures and certain processes at an execution or activity level will be generated for trade compliance reviews.

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PLANNING: Protocol for Auditing (Continued)

LLP suppliers will establish a management system to support requests originated by IBM. The LLP supplier interface will involve the process owner in LLP supplier sub-tier suppliers as required. LLP suppliers will use reasonable effort to support requests within 24 hours.

Reports (formal audits or other IBM internal reviews) will not specifically identify LLP suppliers by name but will refer to them as the "supplier/vendor or any other appropriate designation". However, IBM reserves the right to identify LLP suppliers by their corporate names when necessary and will consult Legal about any circumstances that this may be a requirement, e.g., a trade compliance audit. The IBM Control Tower representative will validate and confirm all facts related to LLP supplier executed processes with them prior to signing or agreeing with the IBM internal reviewer's/inspector's/auditor's results.

Resulting findings, recommendations, and actions directed at the sub-tier supplier should be addressed using the current management system and, if needed, the Change Management Request (CMR) process established in the Statement of Work (SOW). This should be focused on the results with the sub-tier supplier implementing those actions they deem necessary to deliver the expected results.

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PLANNING: Fraud Template

Fraud is any intentional act or omission designed to deceive others and resulting in the victim suffering a loss and/or the perpetrator achieving a gain. Fraud can be categorized into fraudulent financial reporting, misappropriation of assets, and improper or unauthorized expenditures.

Description of Fraud Risk	Typical Controls that should Detect the Fraud Risk	Testing Approach
INVENTORY ACCURACY If processes were not effective to verify inventory, losses may go undetected and financial records could be misstated.	Logistic systems and physical inventory are synchronous Inventory variance evaluations are managed timely and accurately and that inventory adjustments correlate to them Level of inventory losses	Select a judgmental sample of inventory from current inventory (including high value parts) and nonconforming materials (NCM) within the manufacturing, warehouse, and cribs and perform test counts.
INAPPROPRIATE DISBURSEMENTS Lack of controls to validate supplier part costs, quantities, unit and total costs could lead to invalid or inappropriate payments, and diversion of products.	Invoice validation processes	Review supporting documentation for accuracy, and validate that cost is approved by the appropriate levels of management. Validate that purchase order quantity match approved demand. Validate that controls are in place to prevent inappropriate disbursements of funds for purchased goods.
BRIBERY Lack of adequate controls to perform accurate and approved rate fair value analysis for the manufacture could result in unfair bids, inaccurate prices, and kickback to IBM employees	Separation of duties (SOD) should avoid individuals to initiate and conceal irregular activities which could lead to fraud.	Determine if the SOD analysis had been performed in accordance with corporate instructions and all conflicts or potential conflicts had been eliminated Review complaints lodged with the Procurement Ombudsman Validate that fair value analysis are completed according to the IBM Blue Book and approved appropriately.

Note: The probability and impact will vary depending on the organization/business unit that is being audited.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Supply Chain Planning

Objective:

Is the supply chain planning process effective to ensure adequate supply to meet customer demand through appropriate planning, while balancing the objective of supply availability with inventory costs? Is there synchronization of demand uncertainty with supply flexibility through management and accurate planning of lead times? Are there reliable measurements in place to gauge the effectiveness of the process?

Risk/Exposure:

An ineffective inventory planning and measurement process can result in poor customer satisfaction due to missed customer shipments. and non-attainment of business objectives due to excess cost.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Supply Chain Planning

Secondary Test: Inventory Planning and Measurements (SC001.S)

Objective:

To determine if the supply chain planning process is effective to balance between the objective of supply availability with inventory costs, and if there are reliable measurements in place to gauge the effectiveness of the process.

Approach:

Process Objective

To establish annual fall plan and monthly inventory forecast. Define measurements to provide appropriate management visibility to inventory status, and drive action plans to meet customer demands and optimize inventory and costs business objectives.

Process Risk

Achieving business objectives: An ineffective inventory planning and measurement process can result in missed business commitments and excess cost.

Testing Strategy

Obtain copies of all inventory measurements for the twelve months of the audit period.

Review the measurements for appropriate targets, performance trends, and action plans.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Supply Chain Planning

Secondary Test: Supply Demand Planning (SC002.S)

Objective:

To determine if the Supply/Demand Management System is effective and is consistent with demand / supply planning cycle. Is there synchronization of demand uncertainty with supply flexibility through management and accurate planning of lead times?

Approach:

Process Objective

To ensure adequate supply to meet customer demand through appropriate planning, while balancing the objective of supply availability with inventory costs.

Process Risk

Achievement of business objectives: Incorrect demand could result in erroneous supply commitments and excessive cost.

Customer Satisfaction

Inaccurate demand could result in customer commitments not being met.

Testing Strategy

- Is there evidence that all sources of inventory are included in the plan?
- Have inventory exposures been highlighted in the supply decision process?
- Determine if process has flexibility to adjust to demand fluctuation. If so, was agreement obtained for demand adjustments?
- Review demand input and determine if demand is unbiased. How is the basis of the data determined? Is the bias validated? Is there evidence of management agreement?
- Are there key supply/demand process indicators for the planning process? Are they reviewed by management? Are actions taken for missed targets?
- Were commitments to customers updated as a result of supply/demand analysis or management review?

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Supply Chain Execution

Objective:

To determine based on the planning process if orders for parts are properly placed and managed. In addition, to determine if there are adequate controls and reviews for identifying, processing, and resolving excess, surplus and scrap inventory along with timely implementation of engineering changes.

Risk/Exposure:

An ineffective supply chain execution and measurement process can result in poor customer satisfaction due to missed customer shipments. and non-attainment of business objectives due to excess cost.

MANUFACTURING AUDIT PROGRAM #54-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Supply Chain Execution

Primary Test: Material Requirements Planning (SE001.P)

Objective:

To determine that controls over material requirements are generated to help ensure accurate order recommendations.

Approach:

Process Objective

Review Material Requirements Planning (MRP) against the prior forecast cycle, and/or known changes since the last cycle, to determine if any unexpected or obvious deviations exist. To provide feedback of any potential issues found on Demand Planning for review.

Process Risk

Data integrity: Transactions that have been approved for processing may be lost, not completely processed, or inaccurately processed.

Customer satisfaction and achievement of business objectives: Inaccurate MRP data can result in parts shortages, inventory overage, and missed shipments.

Testing Strategy

- Review MRP validation that forecasted demands are assessed to assist Demand Planning to properly load Kinaxis Rapid Response MRP planning.
- Request list of miscellaneous requirements inputs to MRP (e.g., safety stocks), select sample and test to ensure they are properly supported and reviewed.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Supply Chain Execution

Primary Test: Supply Order Management (SE002.P)

Objective:

To determine based on the planning process if orders for parts are properly placed and managed along with timely implementation of engineering changes.

Approach:

Process Objective

Ensure orders for parts are properly placed and managed to meet manufacturing commitments.

Process Risk

Customer satisfaction and achievement of business objectives: Inaccurate supply management can result in parts shortages, inventory overage, missed shipments.

Testing Strategy

- From products that are not meeting schedule commitments, and/or have significant inventory overages or shortages, select one or more months to review. Test for the following:
 - Execution of Material Requirements Planning (MRP) generated requisitions.
 - Actions taken to monitor and control short-lead-time orders.
- Ensure all open purchase orders (including materials identified as critical or short parts) are tracked, reviewed and aged items are resolved timely.
- Review cancellation charges for root cause analysis and proper approvals.
- Request list of unplanned / manual requisitions. Select sample and test for:
 - Reason for manual requisition
 - Properly reviewed
 - Request list of receipt referrals. Select sample and test for:
 - Aging
 - Activity is reviewed and management awareness
 - Timely resolved
 - Review late customer shipments and determine if root cause was related to Supply Order Management process.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Supply Chain Execution

Primary Test: Manual Pulls (SE003.P)

Objective:

To determine if an effective manual pulls process is in place to ensure orders for materials are valid.

Approach:

Process Objective

Ensure execution of manual pulls achieves business objectives.

Process Risk

Customer satisfaction and achievement of business objectives: Untimely materials supply can result in parts shortages, excess inventory costs, or missed shipments and impact cash flow.

Testing Strategy

- Request list of manual pulls for the audit period. Select a sample and test for:
 - Reason for manual pull
 - Appropriate review of manual pull, as required
- List of part numbers pulled during the audit. Select a sample and test for:
 - Pull vs Demand
 - Sample should focus on quarter-end pulls

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Supply Chain Execution

Primary Test: Excess, Surplus, and Scrap (SE004.P)

Objective:

To determine if there are adequate controls for identifying, processing, and resolving excess, surplus and scrap (ES&Z) inventory.

Approach:

Process Objective

To ensure that the process for managing ES&Z inventory is consistent with corporate requirements as defined in CP-10.12.

Process Risk

Lack of appropriate management of inventory overage (ESZ) can result in excess inventory costs.

Testing Strategy

- Determine if controls are in place and supported by a documented process to measure the effectiveness of the site ES&Z management system. Reference CP-10.12.
- Determine that the ES&Z code definition as stated in CP 10.12 are being followed. If not, is the deviation documented and approved by Brand Finance & IBM Worldwide Accounting, with notification to the ESZ process owner?
- Verify that stock location blocking has a valid reason and is reviewed and approved.
- Determine that site purchasing recognizes ES&Z as preferred supply and that vendor terms and conditions provide maximum flexibility to allow optimum ES&Z usage.
- Determine that the site is utilizing all available tools and reports to minimize ES&Z inventory.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Supply Chain Execution

Primary Test: Engineering Change (EC) (SE005.P)

Objective:

To determine that Engineering Changes (EC) are effectively managed and implemented timely, including end of life (EOL) ECs.

Approach:

Process Objective

To analyze and implement ECs to manufacturing line.

Process Risk

Customer satisfaction. Untimely execution of quality ECs could result in defective supply included in customer orders.

Testing Strategy

- Select a sample of quality ECs processed during the audit period. Review documentation to determine the following:
 - The effective date to release date was timely.
 - Aging of ECs was tracked, and actions were taken on aged items.
- From the report of ECs open longer than 60 days select samples to review documentation of status and action plans.
- Review the Bill of Material (BOM) in regard to 'substitutes' and 'use as is' part numbers for accuracy.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Manufacturing Operations

Objective:

Did management have an effective process for planning and fulfilling customer orders in accordance with customer requirements?

Risk/Exposure:

A defective process for manufacturing planning and fulfillment could result in missed customer shipments and excess cost.

MANUFACTURING AUDIT PROGRAM #54-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Manufacturing Operations

Primary Test: Customer Order Management (MO001.P)

Objective:

To verify that customer orders were being fulfilled appropriately.

Approach:

Process Objective

Ensure customer orders passed to manufacturing are complete and accurate to prevent delays in the start to build process.

Process Risk

Data Integrity: Incorrect or incomplete orders accepted by IBM could result in incorrect shipments, inaccurate pricing, noncompliance with contract terms and conditions, and collection issues.

Customer Satisfaction: The consequence of any or all of these factors could lead to increased cost and financial liability (i.e., bad debt write-offs) to IBM, and dissatisfied customers and business partners.

Testing Strategy

- Review the process for communicating customer orders to manufacturing and verify:
 - The appropriate controls were in place to verify data integrity of order transmissions.
 - Adequate process for validating the receipt of the order by manufacturing. Process should include error and exception handling.
- Have metrics been established to track and age invalid customer orders? If so, review metric attainment and action plans where metric was not achieved.
- Verify that backorders were tracked and aged. Assess the level of management visibility to backorders and verify that appropriate actions were taken to fulfill the orders as quickly as possible.
- Test to see if there was a process to monitor and control short-lead-time orders and firm orders in excess of lead-time.
- Request list of late shipments. Based on missed target results select a sample of late shipments and test for the following:
 - Root cause analysis was performed.
 - Action plans implemented to address root cause.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. Inventory and Asset Controls

Objective:

Does management have an effective process to safeguarding assets in inventory both physically and logistically?

Risk/Exposure:

The lack of proper controls over the security of IBM assets can lead to lost or stolen inventory.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. Inventory and Asset Controls

Primary Test: Physical Security (AC001.P)

Objective:

To verify that IBM's assets were secured and only accessible to authorized individuals with a business need.

Approach:

Process Objective

To meet requirements for the protection of IBM owned and consigned parts, machines and physical assets within IBM or at any of the "Supplier" business.

Process Risk

Loss of Assets: Access to materials by unauthorized individuals and failure to account for all assets can result in undetected losses.

Testing Strategy

- Review the latest version of the IBM Corporate Security Policy (ICSP) related to physical and parts security. Test for specific controls related to the various types of security level for designated space.
 - Non-Sensitive Parts Security
 - Sensitive Parts Security
- Obtain a listing of storage locations for sensitive Parts. Review locations to determine if the physical space meets the physical security requirements under C.P 10.13 accordingly.
- Review the latest version of the IBM Corporate Security Policy (ICSP) related to access controls. Test for specific controls requirements.
- Validate that individual IDs are assigned to all Control Assess System (CAS) operating personnel and authorization reviews are performed as required.
- Verify that an effective process is in place to remove employee access in a timely manner.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. Inventory and Asset Controls

Primary Test: Sensitive Parts Management (AC002.P)

Objective:

To determine if asset protection initiatives are attained (the preservation IBM revenue through proper asset controls, parts disposition in a manner preserving IBM's quality reputation, and asset protection decisions provide benefit to the IBM Company).

Approach:

Process Objective

Ensure processes for sensitive parts are adequate to protect them from loss or theft and the loss of sensitive parts are recognized and reported in accordance with CP 10.13 guidelines.

Process Risk

Compliance with Corporate Procedure 10.13 could mitigate the possibility of lost or stolen inventory

Testing Strategy

- Obtain a report of all sensitive parts available at the site during the audit period
 - Select a sample of receipts and disbursements of these part numbers and verify that they can be tracked from point to point required by their Asset Protection Classification (APC).
- Review open APC4 shipment report for timely closure and escalation where required.
- Review open APC4 receipt report for timely closure and escalation when required.
- Obtain a listing of inventory adjustments for parts during the audit period.
 - For lost APC 4 parts were timely identified and reported promptly to management.
 - Verify that all lost parts were accurately entered into the corporate asset tracking system (CATS) per CP10.13 requirements.
 - Verify detection of loss and correction of any out-of-tolerance conditions that may exist and appropriate reporting to Security and Management where variances exist above approved levels in accordance with Corporate financial guidelines.
 - Any unexplained negative inventory variances after investigation are reviewed per local process for the possibility of theft or fraud as documented by the business unit's management approved reconciliation process.
- Ensure inventory counts for sensitive parts are based on CP10.13 requirements.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. Inventory and Asset Controls

Primary Test: Inventory Controls and Verification (AC003.P)

Objective:

To determine if an effective management system was in place to ensure proper accounting of IBM inventory.

Approach:

Process Objective

Validate inventory accuracy within the manufacturing operation.

Process Risk

Asset Loss: If processes were not effective to verify inventory, asset losses may go undetected and financial records could be overstated.

Missed business objectives: Inaccurate inventory records could negatively impact the requirements generation process, which relies on these records to calculate order points and quantities.

Testing Strategy

- Determine if rotating inventory counts (both RICs and 100% counts) were performed and if all inventories were included (e.g., component, work-in-process (WIP), finished goods, and consignment. Test against the following criteria:
 - Request monthly results for the period being audited. Review the results to determine if variance levels were calculated correctly and that the variances were in compliance with parameters.
 - Select a sample of inventory adjustments and test that they were properly analyzed and approved prior to processing.
- Conduct independent inventory counting (book-to-wall, wall-to-book) and verify if warehouse environment is in line with the vendor contract as well as IBM's policy. Please refer to the counting sheet sample "*Template WH Inspection_v0.xlsx*".

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. Inventory and Asset Controls

Primary Test: Inventory Scrap (AC004.P)

Objective:

To determine if the process for final disposition of new and used parts and machines adequately protects IBM assets?

Approach:

Process Objective

Ensure process adequately protects IBM assets throughout the scrapping process.

Process Risk

Loss of Assets: Ineffective controls in the tracking of items scheduled for scrapping and disposal could result in financial loss to IBM and misuse of assets. In addition, IBM's competitiveness could be impacted through the loss of parts scrapped or disposed of that still have value.

Testing Strategy

- Select a sample of scrapped part and test for the following in accordance with CP 10.12.
 - Ensure that all scrapped were properly approved.
 - Determine if cycle time of receipt to scrap (both physically and logistically) is appropriate and meeting targets.
 - Determine (for APC 4 parts) that proper transactions are recorded in the central asset tracking system (CATS) database.
 - Determine if a process exists to ensure all scrapped parts are also retired financially in a timely manner.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. Inventory and Asset Controls

Primary Test: Product and Inventory Returns (AC005.P)

Objective:

To determine if IBM product returns were properly authorized and controlled.

Approach:

Process Objective

To maximize the return on investments in parts and equipment previously owned or shipped and uninstalled through the timely receipt and disposition of these assets as new, equal to new, or scrap.

Process Risk

Loss of Assets: Commodities could be exposed to loss if the management control system does not ensure that returned assets were monitored for complete shipment and return to inventory.

Testing Strategy

- Review the process to determine the processing, approving, and documenting required for Return Parts Miscellaneous Equipment Specifications (RPMES)/ Engineering Change (EC)s and Shipped and Un-installed (S&U) returns, test for the following attributes:
 - Authorizations for returns existed and were approved by S&U Coordinator and/or IBM management prior to the return of the products?
 - Unauthorized returns were reviewed and handled appropriately?
 - Returned product was verified against the system to confirm that the product was previously sold?
 - For S&U returns, credits or replacement parts were promptly processed? Was there a measurement to monitor aging?
 - The customer was notified for resolution when a discrepancy was found with the part number or quantity returned? These products were held apart from available inventory until resolved.
- Obtain a report of all RPMES orders shipped during the audit period. Select a sample to verify compliance to the CP 10.15 procedures.
 - Management process existed and executed to monitor timely reconciliation, closure, escalation, reporting to Security.
 - RPMES shipped but not closed within 90 days are monitored using the IBM Asset Protection Competency Center (APCC) User Tool.
- Obtain a list of all loaned products Joint Engineering Manufacturing Test (JEMT)s and Work-In-Process (WIP) diverts. Select a sample and test compliance to accounting practice AP-INV – COST- Inventorying Cost for New Technology Product and System Brands.

MANUFACTURING AUDIT PROGRAM #54-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 5. System Access and Separation of Duties

Objective:

Are there adequate access and update capability controls and separation of duties to prevent unauthorized changes to programs or fraudulent transaction processing?

Risk/Exposure:

Separation of duties involves the assignment of responsibility for a transaction in a manner such that the duties of one employee automatically provide a cross-check on the work of one or more other employees. The primary purpose of separation of duties is the prevention and prompt detection of errors or irregularities in the performance of assigned responsibilities. Functions are considered to be incompatible from a control standpoint when it is possible for an individual to commit errors or irregularities in the normal course of his duties and to prevent their detection by the system. Individuals with incompatible job responsibilities and system access capabilities are an exposure to IBM.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 5. System Access and Separation of Duties

Secondary Test: System Access Controls (SA001.S)

Objective:

Verify that management controls and authorization of all access to the application and data can be demonstrated with management awareness and removal of such access is processed when there is no longer a business justification (e.g., at separation or change of assignment); a valid business need for privileged userids exists and is current; accountability for shared userids can be established; and that auditable records are maintained to show compliance with IBM security standards.

Approach:

Process Objective

The process objective in IBM's security strategy is to grant or deny access to services and information based on authenticated identity, individual business need, and on broad entitlements.

Process Risk

Data Integrity: Unauthorized persons may gain access to the application and its resources enabling them to enter unauthorized transactions or to make unauthorized changes to data or programs.

Testing Strategy

- Select sample of userids and test for authorization and revalidation.
- Select a sample of users who have left the company and requests to revoke access sent from a manager to the application security administrator. Verify that the userids were removed or disabled in a timely manner.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 5. System Access and Separation of Duties

Secondary Test: Separation of Duties (SA002.S)

Objective:

To verify that no one individual had incompatible responsibilities that would allow the system of internal controls to be circumvented.

Approach:

Process Objective

Provide management with reasonable assurance that no one individual has responsibilities or accesses that would allow employees to misuse or divert company assets.

Process Risk

Failure to establish controls which identify conflicts and failure by management to eliminate or mitigate those conflicts could result in fraud, error, undetected waste or loss of IBM assets.

Testing Strategy

- Review SOD assessment and determine if all elements of aggregate SOD are included. These elements include job responsibilities, physical access to inventory and all critical application access.
- Review SOD assessments to determine conflicts are appropriately identified including key areas such as ES&Z, customer order management, inventory controls and verification, and inventory scrap.
- Review employee SOD analysis output and determine if conflicts were identified and actions taken to resolve the conflict.
- If conflicts could not be resolved, ensure that the secondary controls were implemented which effectively mitigate the risk of asset loss.

MANUFACTURING AUDIT PROGRAM #54-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 6. Customer Solution Center

Objective:

To determine whether solutions delivered were in accordance with commitments, that internal and external resources were monitored and utilized effectively, and revenue recognition.

Risk/Exposure:

Because of the complexity or nature of Customer Solution Center (CSC) projects, the execution phase requires focus on IBM performance, versus the contract commitments and the customer expectations. The lack of project follow-up and of monitoring for subcontracted services, and the failure to effectively use or re-allocate resources could jeopardize IBM's ability to fulfill the contractual commitments and result in cost overruns, impacting profit.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 6. Customer Solution Center

Primary Test: Deliverables (CS001.P)

Objective:

To determine if management checkpoints and controls were implemented for statement of work deliverables to ensure they were on schedule and within budgeted cost.

Approach:

Review the statement of work to understand the deliverables and request the universe of all deliverables. Review all or select a sample of deliverables to test.

The test should assess the effectiveness and timeliness of management checkpoints and compliance to statement of work requirements across the service delivery process. Verify the following:

- The statement of work (SOW) was clear and well defined and containing appropriate business and legal clauses and basic deliverables such as quantity, timeliness, and inventory controls if required.
- If the statement of work contained Service Level Agreement (SLA) targets, were they attained? If not, were root cause analyses conducted? Were action plans put into place, scheduled, and reviewed until corrective solutions were completed in order to meet SLA targets?
- Acceptance tests as required by the statement of work terms were performed on schedule and formally agreed by the customer.
- Checkpoints existed in the process at the completion of key work items. Each checkpoint should provide a status of the project against an established level of expectation, and reporting to the client should be consistent with the statement of work requirements.
- If checkpoints and expectations had not been met, perform the following test: Were there action plans to correct deviations from the plan with a degree of efficiency that did not increase costs? In those instances where the deviations increased costs, were the additional costs properly reviewed and approved by the customer (if the additional costs can be charged to the customer) and IBM? For unanticipated situations, ensure that management had developed additional checkpoints and effective controls to handle future occurrences.
- Assess the root cause of instances where deliverables have not been met, is this due to solution design or delivery.
- Assess for effective control of transition activities.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 6. Customer Solution Center

Primary Test: Asset Management (CS002.P)

Objective:

Ensure that ownership of all assets which were part of the statement of work was clearly defined and the responsibility for management and control of the assets was identified. In addition, ensure that all assets (owned or leased) are managed in accordance with applicable statement of work terms.

Note: Assets can include hardware, licensed programs, and other property as identified in the statement of work.

Approach:

Review the statement of work and its associated schedules to understand the statement of work requirements for all asset management activities. This should include IBM owned or leased assets and customer owned or leased assets.

Were all HW assets listed and reconciled to the statement of work? Were all SW programs listed and reconciled to the statement of work?

The most recent inventory should be compared against the appropriate statement of work schedule for accuracy and current status.

Documentation should exist indicating the location of all assets associated with this statement of work. Specifically, the list should include where the asset(s) are located, i.e., on IBM premises, at the customer site, or suppliers' premises.

Hardware

- Obtain a copy of the latest hardware inventories and verify that at least annual physical inventories had been performed, including inventory of IBM and customer assets, as defined in the statement of work. Inventories should include OEM hardware.
- Inventory reconciliations were reviewed and approved by IBM management.
- Sample a portion of the inventory to determine if the hardware could be located and was consistent with the inventory records ('book-to-floor') and select some hardware from the premises and verify that the equipment matched the inventory records ('floor-to-book').
- Check that we are meeting statement of work responsibilities regarding assets.
- Review monthly compliance testing results for out of line situations.
- Initial inventory was carried out according to the terms of the statement of work.
- An appropriate asset management tool is used to record the assets.
- Appropriate separation of duties existed between personnel performing inventories and personnel responsible for updating control files. Review list of people with access to inventory change facility.
- Is there a process to cover all asset management activities: new, acquired, reassigned, lost, retired, disposed. Are all capital expenditures appropriately planned and approved?

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 6. Customer Solution Center

Primary Test: Asset Management (CS002.P)

(Continued)

- Billable asset charges are clearly identified for billing to the customer in accordance with the terms of the statement of work.
- IBM owned assets are appropriately listed in the authorized corporate inventory system (e-AMT).
- Customer-owned or leased assets are managed and controlled. The customer should be informed of any loss of customer assets.
- Review the project office procedure for procurement of OEM products and verify that documented safety standards for OEM products were required.

Software/License Programs

- Is IBM performing according to statement of work requirements?
- Is there a current software inventory for the statement of work, agreed by the customer?
- Is there a process to cover software asset management activities: adds, deletes, changes?

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 6. Customer Solution Center

Primary Test: Product Safety (CS003.P)

Objective:

Determine if a process was in place to ensure all OEM products and modified IBM products had Product Safety Review Board (PSRB) approval, prior to delivery or prior to export. (Note: approval is at product, not customer contract, level).

Determine if products requiring product safety reviews were being identified as such in the solution design development process and that approval was obtained prior to commitment to the customer and vendor.

Approach:

- Based on sampling methodology, ensure some proposals selected for auditing included solutions containing OEM products or modified IBM products.
- Verify that appropriate PSRB approval for the product has been obtained prior to contracting with the supplier.
- Review the final contract to ensure that equipment provided or integrated in the solution had PSRB approval.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 6. Customer Solution Center

Primary Test: Project Control and Contract Changes (CS004.P)

Objective:

Ensure that adequate, effective, and timely management controls exist to meet the business and customer specifications for all service requests, projects; both in-scope and out-of-scope of the contract, and changes to the services or contract terms and conditions.

Approach:

Review the process for implementing projects and making changes to the statement of work and determine if the service request or statement of work change management process is adequate, effective, and timely.

- Does a target completion date exist?
- Does it capture all requests and requirements for change?
- Does it validate the business and/or customer justification for each individual change?
- Does it require that all justifications for change are communicated and negotiated with the customer for approval prior to implementation?
- Does it include a management and customer review and approval phase to ensure that each change is approved at a level commensurate with the magnitude of the change?
- Does it require the sizing and costing of all changes to ensure that the full scope of each change is adequately considered prior to approval?
- Is there a tracking mechanism to resolve open items?

Select a sample of projects, service requests, and statement of work changes not identified in the initial statement of work and determine if the request and change processes were followed. Verify that all changes are incorporated into the cost and schedule reporting systems.

For all projects, compare committed costs against billed costs to determine if cost objectives were met. Also, review committed costs to determine if any major costs were incurred which were not planned for in the original budget. For additional work outside the scope of the statement of work, ensure that the agreement for the terms of that work is obtained prior to starting the work.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 7. Property Control

Objective:

Determine whether there were controls in place to ensure that assets were physically inventoried in accordance with corporate requirements, and actions taken when assets were declared missing.

Risk/Exposure:

Failure to control the physical verification of fixed assets would impact the ability to detect losses and may impact the integrity of IBM's financial statements.

MANUFACTURING

AUDIT PROGRAM #54-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 7. Property Control

Primary Test: Physical Inventory and System Integrity (PC001.P)

Objective:

To evaluate whether the asset tracking system records accurately reflect the:

- Physical existence and location of assets (internal and at Suppliers),
- Asset owners have been assigned, and
- That updates to the asset records were performed in a timely manner or the asset was returned, if applicable.

To evaluate controls when assets were physically inventoried with an emphasis on actions taken when assets were declared missing.

Approach:

1. Select a sample of assets from the property control system (e.g., e-AMT), that are high risk (e.g. notebook computers, capitalized value over \$5,000).

- Review the information on the system and verify the assets are found and asset record information is stated correctly.

Select a sample of assets from each of the following categories:

- Assets assigned to asset coordinator instead of an individual (for example functional or warehouse locations, or departments). In some cases, these asset coordinators could be staging areas for handling assets.
- Assets currently active on the asset record but have invalid location or department information on the asset record.
- Assets that were assigned to users who have either separated from IBM or moved internally which have been re-assigned in the property control system to the manager responsible. If assets are not properly re-assigned, assets may be unlocated which have not been detected.
- Assets assigned to departments which have had significant write-offs of unlocated assets (both quantities and values).
- Assets loaned to Suppliers

Review the physical inventory process and determine if the procedures adequately define the physical verification process and all assets were captured as required by procedures

Testing Strategy

- For assets assigned to an asset coordinator focus on the transfer dates to determine if the transfer occurred in the property control system. Annual physical inventories are generally random sample counts. Unless these asset coordinators are effectively managing their assets, unrecorded transfers and loss of assets could occur and remain undetected. Test for the following:
 - The asset record is accurate. (e.g., assigned to an active user, address, bldg, loc, description matches, etc.)
 - Procedures were in place to tag assets. If tags were damaged or lost, was there a process for re issuing tags?

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 7. Property Control

Primary Test: Physical Inventory and System Integrity (PC001.P)

(Continued)

- Assets Loaned to Suppliers must be adequately controlled, must be listed in e-AMT and the employee is responsible for these assets in the same manner as assets located at IBM. Test for the following:
 - The eAMT asset owner maintained an accurate inventory list of assets loaned to Suppliers.
 - A current Equipment and Program Loan Agreement (EPLA) Supplement is in place (if required).
 - The asset record is accurate (e.g., assigned to an active user, address, bldg, loc, description matches, etc.). Verify the location in eAMT matches the location of the supplier.
 - Loaned IBM assets were recovered from the supplier when no longer required and the eAMT record and EPLA Supplement were updated.
- Physical Verification Process. Test for the following:
 - Physical or paper inventory of assets was performed as required.
 - Management had visibility to the results.
 - Assets were properly returned, and the asset record was updated.

Reverse Check

To evaluate asset tracking system record update controls when assets have been physically received at the locations.

2. Tour the location and select a sample of inventoriable assets that are high risk and determine if the information is stated correctly in the property control system. Inquire about warehouse and storage space where idle assets are held.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 8. Measurements

Objective:

Were measurements in place to provide management visibility to supplier performance against the requirements specified in the contracts? Were controls over the Quality Management System adequate to verify that IBM products produced by an outsourced manufacturer were able to meet or exceed customer expectations? Was the vendor committed to providing quality products in terms of reliability, and performance?

Risk/Exposure:

Failure of suppliers to meet established performance targets decreases IBM's ability to provide high-quality, low-cost products and services to our customers, and could result in increased support costs (call center, support staff, etc.); increased error rates, resulting in lower productivity; decreased product sales; increased warranty costs; decreased customer satisfaction; and a negative corporate image. Additionally, the absence of a good internal measurement system creates an environment for fraud to occur and can lead to excessive expenditures.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 8. Measurements

Primary Test: Measurements (MM001.P)

Objective:

To determine if measurements were in place which provide management visibility to supplier performance against the requirements specified in the contract and scope of work. To validate the Quality Management System ensures only quality products were being delivered to external customers.

Approach:

Process Objective

Provide management visibility to supplier performance against the requirements specified in the contracts and ensure product quality meets or exceeds IBM standards or expectations to ensure customer satisfaction.

Process Risk

Failure of suppliers to meet established performance targets decreases IBM's ability to provide high-quality, low-cost products and services to our customers. Additionally, the absence of a good internal measurement system creates an environment for fraud to occur and can lead to excessive expenditures.

If the Quality Management System did not have the controls in place to verify that outsourced manufacturers were providing quality IBM products, this could result in: increased support costs (call center, support staff, etc.); increased error rates resulting in lower productivity; decreased product sales; increased warranty costs; decreased customer satisfaction; and a negative corporate image.

Testing Strategy

- Review line management measurements to determine if supplier performance criteria for procured goods and services were monitored and reported to management. Evaluate the management process for determining that performance measurements were achieved.
 - Had measurements been established, including reporting frequency and management review of reports to determine that required deliverables specified in the contracts were being received?
 - Were these measurements consistent with contract terms? Did measurements focus on timeliness, customer satisfaction and quality?
 - Was the vendor providing timely and accurate measurements to IBM as required in the contract? Was measurement source data provided by the vendor? If so, was there a process for verifying the accuracy of this source data?
 - Were supplier audits conducted and, if necessary, were corrective actions taken based on the results?
 - When supplier performance targets were not met, were there documented action plans developed?
- Had IBM communicated its product quality, reliability, and performance objectives to the vendor? Was the vendor measured against these objectives?
 - When these objectives were not met, were actions plans developed, assigned owners, and given due dates for resolution? Were quality problems clearly traceable from problem identification to resolution?
 - Was there an established and routine process for communicating quality concerns and issues to the vendor?
 - If on-site vendor reviews were performed, was there a process for tracking action items from these reviews to resolution?
 - Was there a process for monitoring the material suppliers used by the vendor?

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 9. Supplier Payment Controls

Objective:

Were the controls for supplier payments, credits, and price reconciliation activities adequate to ensure the appropriate placement of checks and balances throughout the process? Was there sufficient management involvement within the key control points and activities? Were the monitoring and control functions surfacing exception conditions and initiating appropriate corrective action?

Risk/Exposure:

Fraud may be perpetrated, and cash flow may be diminished due to inaccurate supplier payments and reduced receipts. In addition, if established guidelines are not enforced, customer satisfaction could diminish, and IBM could be exposed to litigation associated with questionable business practices.

Note: To be used in manufacturing audits where Production POs are placed directly in the order system.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 9. Supplier Payment Controls

Primary Test: PO Processing (SP001.P)

Note: To be used in manufacturing audits where Production POs are placed directly in the order system.

Objective:

To determine if controls over the purchase order (PO) process were effective to ensure appropriate commitment of funds, commitments were made to valid suppliers, purchases were valid, approvals were obtained, and IBM's interests were protected.

Approach:

Process Objective

Ensure POs are properly authorized for the commitment of funds, the correct supplier is selected, the PO references the correct Ts&Cs and correct pricing is used.

Process Risk

Failure to properly execute POs could lead to unauthorized commitments, overpayment for goods or services, lack of legal protection, and/or early or late payments to the supplier.

Testing Strategy

Using data analysis and risk based auditing techniques (see Sampling) select a sample of purchase orders.

1. Was there a valid business purpose? If applicable, was the appropriate commodity code used?
2. Was the PO committed to a valid supplier?
3. Were proper approvals obtained prior to commitment to a supplier?
4. Was the price equal to the accepted arrangement (contract, price list, quote, or cost estimate)?
5. Did the PO specify terms for invoicing, payment, and delivery?
6. Where applicable, did the PO commitment amount equal the actual payment(s) amount?
7. Was the PO issue date prior to the accounts payable invoice date? If not, was the PO coded as a bypass and was there a valid reason for this exception?
8. Was the purchase order properly coded? Was there a process to validate the accuracy of source codes entered into the PO system? If inaccurate sources codes were identified, were they corrected in a timely manner and were actions taken as appropriate?
9. Were all changes processed based on alterations properly?
10. Were cancellation charges properly approved?
11. Were any premium charges such as premium freight and overtime properly approved?

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 9. Supplier Payment Controls

Primary Test: PO Processing (SP001.P)

(Continued)

12. Were delegations and/or clip levels circumvented by placing multiple purchase orders to the same suppliers?
13. Obtain the Requisition Cycle Time Report to determine POs with large cycle time between requisition and PO placement. (Requisition processing more than 20 days). Determine if the requisition cycle time was greater than a specified local target (i.e., 10 days). Determine the causes for the delays.
14. If unit-less purchase orders were used (the unit designation in SAP is 'UL'), request a list and select a sample for testing. Samples should focus on high-dollar value, purchases that were paid with multiple invoices, and purchases with descriptions that could be itemized into individual units. For each PO determine if:
 - a. The use of the UL designation was appropriate. Could the purchase have been defined in units?
 - b. Management and Procurement were able to demonstrate exactly what was purchased.
 - c. Management implemented regular UL PO reviews. Review the methodology for selecting UL POs for this testing to ensure complete coverage of UL activity.
 - d. Controls were in place to verify that IBM received and was billed for only what was ordered through proper procurement channels. (The negative confirmation process with accounts payable is not sufficient since this will go to the requester.)
 - e. The measurements and use of UL designations complied with Procurement requirements.

MANUFACTURING

AUDIT PROGRAM #54-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 9. Supplier Payment Controls

Primary Test: Supplier Payment Controls (SP002.P)

Objective:

To determine that payments to the outsourced manufacturer were for goods and services that had been received, properly approved, and in compliance with the contract terms and conditions.

Approach:

Process Objective

Ensure payments to the supplier are valid, accurate and timely per the contract terms and conditions.

Process Risk

Fraud may be perpetrated, and cash flow may be diminished due to inaccurate supplier payments and reduced receipts. In addition, if established guidelines are not enforced, customer satisfaction could diminish, and IBM could be exposed to litigation associated with questionable business practices.

Testing Strategy

- Select a sample of supplier payments against the contracts and test against the following criteria:
 - Was there an independent verification of receipt of goods (production related) or services performed (including all non-production related payments)? Was documentation of this verification available to support payments made?
 - Were IBM liabilities for inventory maintained at vendor locations consistent with the contract terms and conditions?
 - Did the invoice price and quantity match commitment amounts from the contract and the scope of work?
 - Did payments exceed commitment amounts in the contract or purchase orders? Was there a method to track and reconcile payment activity against the contract?
 - If any invoices contained hours worked, was there a process to verify hours worked with hours billed by the vendor?
- Review the process for requesting and receiving credits from the vendor(s) and assess controls to verify that claims for shortages, returned goods, or deficient services were filed and resolved promptly, and that credit notes were received from the supplier promptly. NOTE: This would include any credits for volumes committed to, but not met by the supplier.
- Test the process used to ensure that actual supplier prices were consistent with prices authorized by IBM procurement.

MANUFACTURING AUDIT PROGRAM #54-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 10. Healthy and Safety Programs

Objective:

Do procedures and controls over health and safety requirements to verify that these requirements are fulfilled?

Risk/Exposure:

Failure to properly fulfill healthy and safety requirements can lead to financial loss and damage to the IBM brand.

MANUFACTURING AUDIT PROGRAM #54-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 10. Healthy and Safety Programs

Secondary Test: Equipment / Process Installation (HS001.S)

Objective:

The introduction of new equipment and processes has the potential to introduce risks to those installing, using, and maintaining the equipment. The use of an equipment review process will identify potential risks and enable appropriate controls to be established.

Approach:

1. Is a process established to identify and minimize potential risks posed by new equipment/processes?
2. Are risks eliminated and/or reduced in accordance with the hierarchy of control principles via: Elimination, Substitution, Enclosure/segregation, Engineering controls, administrative controls, Personal Protective Equipment (PPE)?
3. Is a program of appropriate, ongoing of equipment/processes maintenance established?
4. Is a provision made for suitable instruction/training of operators and maintenance staff as appropriate?
5. Are all above processes and requirements adequately documented?

MANUFACTURING AUDIT PROGRAM #54-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 10. Healthy and Safety Programs

Secondary Test: Potentially Hazardous Work Activities (HS002.S)

Reference Information

IBM Global Well-being Practices: Potentially Hazardous Work Activities

Objective:

Certain activities can result in potentially hazardous exposures and conditions. Controls are required to prevent injury or ill health. Activities include, but are not limited to, those involving ionizing and non-ionizing radiation, noise, chemicals, manufacturing ergonomics, electrical hazards, process maintenance operations, etc.

Approach:

1. Are assessments carried out to identify health and safety risks in connection with potentially hazardous activities?
2. Are risks eliminated and or reduced in accordance with the hierarchy of control principles via: Elimination, Substitution, Enclosure/segregation, Engineering controls, Administrative controls, personal protection equipment (PPE)
3. Are preventative maintenance and or protective control measures documented with clearly assigned responsibilities?
4. Are health surveillance programs in place?
5. Are employees provided with information, awareness, education or training in accordance with the risks identified?
6. Is the IBM operation in compliance with the limits set by Integrated Health Services for Noise, Chemical Exposure and personal exposures?

Lockout/Tagout

Review the lockout/tagout (LO/TO) program and related documentation and validate the following:

1. Verify that procedures address contractors performing LO/TO on IBM premises.
2. Test that all individuals authorized to perform LO/TO had been appropriately trained. Validate that certification of training was in place for authorized, affected, and other employees.
3. Test that annual reviews of a representative sample of LO/TO procedures had been conducted and documented.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 10. Healthy and Safety Programs

Secondary Test: Potentially Hazardous Work Activities (HS002.S)

(Continued)

4. During a tour of selected areas, select a sample of equipment and test that an accurate and complete LO/TO procedure existed, including references to the following:
 - a. Use of unique locks and tags
 - b. Preparation for shutdown
 - c. Lock and tag application
 - d. Release of stored energy
 - e. Isolation and verification
5. During the sampling tour, observe LO/TO work and conduct an employee interview to test that procedures and program requirements were being properly understood and executed. Also, test that individuals performing LO/TO work had been authorized.

Electrical Safety

Review the electrical safety program and related documentation and validate the following:

1. Procedures existed which defined electrical safe work practices for hazardous voltages.
2. Individuals qualified to complete various types of electrical work had been identified, appropriately trained, and training documented.
3. A process to confirm that work with hazardous electrical energy, other than troubleshooting, diagnostics, and calibration activities, meets the established criteria, and includes documented assessments of the rationale for performing work on energized equipment for other than troubleshooting, diagnostics and calibration activities.
4. For high voltage work, switching and grounding procedures had been used by employees servicing generation, transmission, and distribution systems.
5. Electrical protective equipment was stored, maintained, tested, and used consistent with applicable standards.
6. Electrical disconnects were provided to safely disconnect and isolate equipment from its power source.
7. Extension cords were only used for temporary purposes.
8. Ground Fault Circuit Interrupter (GFCI)s was required on circuits servicing outlets in damp, wet, or outdoor locations, and in any other location where individuals using electrical equipment could become well grounded. GFCIs were tested according to manufacturers' recommended frequency.

MANUFACTURING AUDIT PROGRAM #54-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 10. Healthy and Safety Programs

Secondary Test: Potentially Hazardous Work Activities (HS002.S)

(Continued)

Radiation Safety Program

Review the radiation safety program and related documentation and validate the following:

1. Radiation and laser safety officers had been designated.
2. Does a process exist to ensure radiation sources, including equipment and materials, had been approved by the appropriate safety officer prior to their purchase, installation, movement, or reassembly.
3. Inventories had been maintained for class 3B and 4 lasers, ionizing radiation sources, and significant non-ionizing radiation sources.
4. Periodic surveys had been conducted to measure equipment leakage and ensure that equipment remains within established control parameters. Instruments used for these surveys were calibrated.
5. Radiation sources, including equipment and materials, had been properly disposed.
6. Employee training had been completed and documented.
7. Employees with potential exposure to ionizing radiation sources had been identified and employee exposure monitoring had been completed, if appropriate. Results must be communicated to employees.
8. Eye exams had been completed for all employees with potential exposure to class 3B and 4 lasers radiation.
9. Licenses and permits were current, where applicable.
10. Inspect a sample of high-risk radiation source installations, focusing on ionizing radiation and class 3B and 4 lasers. Test the following:
 - a. Location Inventory was accurate.
 - b. Appropriate confinement and intrusion controls had been implemented and were functioning as intended.
 - c. Equipment safety devices, such as interlocks and covers, were labeled correctly and functioning properly.
 - d. Required personal protective equipment was available and in good condition.
 - e. Employees using the equipment had been trained.

MANUFACTURING AUDIT PROGRAM #54-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 10. Healthy and Safety Programs

Secondary Test: Manual/Materials Handling (HS003.S)

Objective:

To determine if effective materials handling processes had been implemented to protect employees in compliance with IBM and regulatory requirements.

Reference

IBM Global Well-Being Practices: Manual/Materials Handling

Approach:

Review the materials handling safety program and related documentation.

1. Is general information on manual/material handling made available to employees?
2. Are activities where manual/materials handling are foreseeable assessed for risk? Control measures must be specified where needed including avoidance where possible, mechanical assistance and working methods to minimize risks.
3. Are formal manual/materials handling training programs established, delivered, and documented where needed in accordance with risk assessments? Appropriate training for mechanical lifts must be provided.
4. Is consideration given to individual risk factors where needed including disabilities, medical conditions, pregnancy?

MANUFACTURING AUDIT PROGRAM #54-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 11. Export Compliance

Objective:

To confirm that products, technology to be exported were properly reviewed for export classification prior to export and the appropriate licenses were obtained. Confirm that appropriate US deemed export and re-export authorizations, and related internal control programs are in place anytime source code or technology is communicated, exposed, released, or transferred to or in the presence of foreign persons working in countries in which they are not citizens or legal residents.

MANUFACTURING

AUDIT PROGRAM #54-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 11. Export Compliance

Primary Test: Regulatory Licenses Requirements (EC001.P)

Note 1: Applicable to organizations within IBM Systems if the organizations are involved in the design, development, release, transfer, or utilization of source code or technology or export of controlled technology products.

Note 2: When a foreign person in the United States has access or exposure of any kind to source code or technology, such access or exposure is deemed to be equivalent to a direct transfer of that source code or technology to the foreign person's home country, i.e., a deemed export.

A deemed re-export occurs when US origin source code or technology is transferred or exposed to anyone who is not a citizen or permanent resident of the country they live and work in, except persons in those countries who are citizens or permanent residents of the United States.

Objective:

To confirm that products, technology to be exported were properly reviewed for export classification prior to export and the appropriate licenses were obtained. Confirm that appropriate US deemed export and re-export authorizations, and related internal control programs are in place anytime source code or technology is communicated, exposed, released, or transferred to or in the presence of foreign persons working in countries in which they are not citizens or legal residents.

Approach:

Review the location's export shipments for the audit period and determine if the export classifications documented for those items were obtained from IBM authorized sources (e.g., Export Regulation Office (ERO) Hardware Look Up Tool, CCCS) and the procedures for identifying foreign persons who are employed at the audited location for determining any US export compliance requirements that apply to those person's IBM job assignments.

- If licenses were required, determine whether licenses were obtained by IBM prior to the export of the items especially for High Performance Computing (HPC).
- For invoices that were not generated through SAP system, were there documented processes for ensuring export classification were reviewed and obtained from the appropriate sources prior to the exportation of the goods by the IBM manufacturing location?
- A review was carried out for all source code or technology that is required for a foreign person employed by IBM who may have access to while performing their IBM job assignment. This includes (but is not limited to) access to output from or data residing on servers, team rooms, databases, electronic libraries, and any other on-line collaborative environment (e.g., wikis) that the foreign person will be granted access to while working for IBM.
- Particular emphasis needs to be placed on controls for foreign persons from any of the sanctioned countries (i.e. Cuba, Iran, North Korea, Sudan, Syria, and the Crimea, Donetsk, Luhansk, Kherson and Zaporizhia (Zaporizhzhia) region of Ukraine).

MANUFACTURING AUDIT PROGRAM #54-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 12. Import Compliance

Objective:

To validate that manufacturing locations have implemented controls to ensure compliance with import requirements by providing accurate and complete information for used goods, Country of Origin (CoO), description of goods and value of goods.

MANUFACTURING AUDIT PROGRAM #54-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 12. Import Compliance

Primary Test: Import Risk Management (IC001.P)

Note: Refer to the Import Compliance Office (ICO) website for current information with regards to the Import compliance requirements.

Objective:

To validate that manufacturing locations have implemented controls to ensure compliance with import requirements by providing accurate and complete information for used goods, Country of Origin (CoO), description of goods and value of goods.

Approach:

Select a sample of invoices representing a variety of production and non-production shipments, including Customer Solutions Center (CSC) activities and ad hoc shipments to determine if invoices have complete and accurate information such as the description of goods (including quantity), value of goods (ad hoc and CSC shipments only) and Country of Origin (CoO).

- Are processes in place at the manufacturing location to ensure adherence to country-specific requirements for used, refurbished, reconditioned equipment or parts?
- Were Shipping Procedural Instructions (SPI) updates reviewed to ensure compliance with any country-unique country of origin regulations?
- Did the manufacturing location ensure the CoO on articles and immediate containers conformed to IBM CoO Marking Instructions?
- How is a sourcing change to the “essential character” of an article detected and reviewed for its impact on CoO determination?
- How is CoO information input into the system? Were updates made as appropriate?
- For all outbound shipments where manual invoices were used, was there a process to ensure that the invoice descriptions were clear, accurate and consistent between the contents of the physical shipment and paperwork so that an accurate classification decision could be made by the importing country?
- Were controls effective to ensure accurate weights, and dimensions provided by Manufacturing, were properly reflected on shipping documentation and labels for non-production, or ad hoc shipments (manual invoices) that facilitate accurate customs declarations?
- If proforma invoices are permitted, were controls effective to ensure that proforma invoice information matched the billing invoice?
- For non-production and ad hoc shipments where manual invoices were used, did the location manufacturing goods have a process to ensure the value provided for commercial invoices met the IBM valuation requirements?

MANUFACTURING

AUDIT PROGRAM #54-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 13. Import Compliance

Primary Test: Import Risk Management (IC001.P)

(Continued)

- If applicable, for Customer Solution Centre (CSC) shipments, ensure that the value assigned to the goods reflects both the consigned hardware received (if consignment in place) and the price charged to the importing entity (inter-company price if sold to inter-company or customer price if sold directly to customer). CSC services would be included when charges are related to the goods.
- Is it evident that the consignment value properly accounted for all elements included the value-add performed.

[Note: Consignment refers to goods for which the ownership does not follow the physical location (ownership of the goods does not transfer.) So, if the CSC physically handle goods owned by another entity, the shipment of those goods cross border will require consideration of values not part of the price of the goods.]